

# Ambient-Sensing Night Purifier

Innovation dossier

## 1. What is it?

A concept for the real 'Noise / motor sound (weak real signal)' friction, built with the AMBIENT design move - explicit interaction -> invisible environmental behaviour.

## 2. Physical architecture

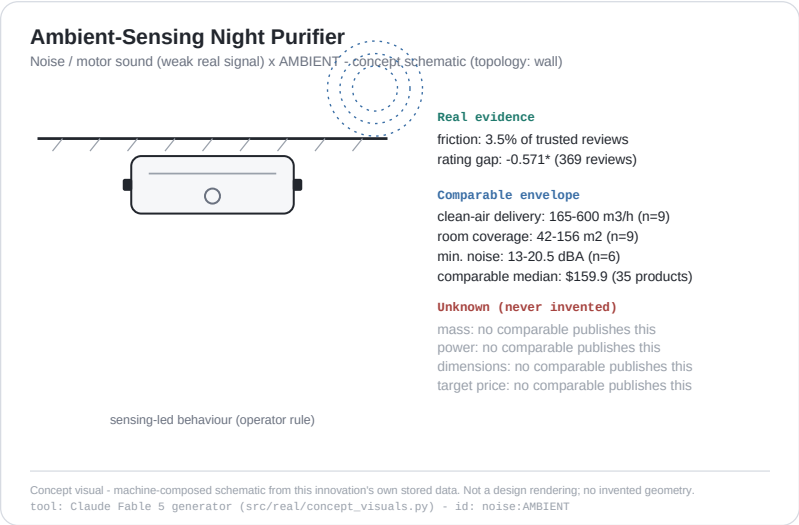


Figure: [Concept visual](#)

Machine-composed schematic from the innovation's stored fields only (operator rule, friction stats, comparable envelope). Not a design rendering; a Fable render or drawing would also be CONCEPT\_VISUAL, never a digital model or physical prototype.

artifact visual-noise:AMBIENT - /concept-visuals/noise\_AMBIENT.svg

- Product archetype**
- room scale:** Unknown - no real data in this pipeline for this field.
  - mobility:** Unknown - no real data in this pipeline for this field.
  - size class:** Unknown - no real data in this pipeline for this field.
  - power class:** Unknown - no real data in this pipeline for this field.
  - noise ambition:** Unknown - no real data in this pipeline for this field.
  - maintenance burden:** Unknown - no real data in this pipeline for this field.
  - filter service dependency:** Unknown - no real data in this pipeline for this field.
  - connected:** Likely - operator AMBIENT (invisible environmental behaviour) requires sensing.
  - form factor:** wall
- Design rule - each populated field follows from the operator's own fixed definition, never from evidence

**System architecture**

Unknown - no system-architecture dataset exists; the archetype above is a design rule from the operator's own definition.

**Method choice**

**AMBIENT - Explicit interaction -> invisible environmental behaviour.**

Method choice - authored design vocabulary

3. Why it exists

Reality Observed

3.5% of trusted reviews (369 reviews across 77 real products) carry this friction, with a -0.571\* average rating gap. Signal: Motor/fan noise is a recurring complaint, now backed by peer-reviewed evidence it drives real under-use.  
Households whose real reviews carry this friction: 369 reviews across 77 real products (69.6% verified purchase, 2004-10-14-2023-08-08).  
*Unknown - no persona/behavioural dataset exists in this pipeline; only the review evidence above is real.*

Transformation Method choice

Method choice, not evidence: the authored operator table assigns AMBIENT - "Explicit interaction -> invisible environmental behaviour." - to this friction. Analyst design judgment, applied deterministically.

Product consequence Research tension

Optimize for hours-actually-run under real noise tolerance, not peak lab CADR. Assumption challenged: "Clean-air delivery rate (CADR) is the primary performance basis." -> What if performance were reported as a loss-rate curve or realized exposure reduction, not a lab CADR number?

4. Evidence and lineage

Parent paths

**tension:T1 - Filtration vs. noise-driven adherence** Tension

test: Falsified when one design point achieves RP-01's measured primary-room particulate reduction at a sound level that removes noise as a named reason for running the device less (RP-05) - both halves are stored, quantified findings.  
derived possibilities: noise:AMBIENT, noise:TEMPORAL\_SHIFT, noise:CROSS\_CATEGORY\_TRANSFER, value\_effectiveness:CONCENTRATE, value\_effectiveness:CROSS\_CATEGORY\_TRANSFER, value\_effectiveness:PERSONALISE

**tension:T2 - Constant vs. automatic operation** Tension

test: Falsified when an auto-mode arm reaches the lower bound of RP-07's constant-mode reduction interval while keeping auto's runtime/noise advantage - the two stored confidence intervals would then overlap and the trade-off dissolves.  
derived possibilities: noise:AMBIENT, noise:TEMPORAL\_SHIFT, noise:CROSS\_CATEGORY\_TRANSFER, value\_effectiveness:CONCENTRATE, value\_effectiveness:CROSS\_CATEGORY\_TRANSFER, value\_effectiveness:PERSONALISE

**tension:T3 - CADR theory vs. real-home performance** Tension

test: Falsified when a CADR-derived sizing prediction matches a source-strength-adjusted measured loss rate within the fan-speed band RP-06 reports, in a home with the central air-handling configuration RP-01 identifies as decisive.  
derived possibilities: noise:AMBIENT, noise:TEMPORAL\_SHIFT, noise:CROSS\_CATEGORY\_TRANSFER, value\_effectiveness:CONCENTRATE, value\_effectiveness:CROSS\_CATEGORY\_TRANSFER, value\_effectiveness:PERSONALISE

**tension:T6 - Pollutant reduction vs. clinical outcome certainty** Tension

test: Resolved when an independent replication of RP-08 reproduces its clinical effect in a cohort whose particulate reduction matches RP-03's measured range - RP-08's own record names independent replication as the missing step. A failed replication resolves it the other way.  
derived possibilities: noise:AMBIENT, noise:TEMPORAL\_SHIFT, noise:CROSS\_CATEGORY\_TRANSFER

**assumption:A5 - Clean-air delivery rate (CADR) is the primary performance basis.**

Assumption to test

test: Challenged when a certification body in the trend corpus adopts a loss-rate or realized-exposure metric alongside CADR - concretely, when TC-R03 (AHAM Verifide, the one record whose theme tags contain 'cadr': ['certification', 'cadr', 'iaq\_standards']) stops centring on it, or TC-R02 (ENERGY STAR, themes [regulation', 'energy\_efficiency', 'iaq\_standards'], CADR present only in its prose scope note) adds a non-CADR performance theme tag. Fragile baseline: both documents carry published\_date null, so change detection has no dated anchor.  
derived possibilities: noise:AMBIENT, noise:TEMPORAL\_SHIFT, noise:CROSS\_CATEGORY\_TRANSFER, value\_effectiveness:CONCENTRATE, value\_effectiveness:CROSS\_CATEGORY\_TRANSFER, value\_effectiveness:PERSONALISE

**assumption:A6 - Continuous, constant operation is the default expectation.**

Assumption to test

test: Challenged when connected+adaptive share exceeds 25% of the corpus (currently 23/237 = 9.7%), or when an independent trial reproduces RP-07's constant-mode filtration under auto operation.  
derived possibilities: noise:AMBIENT, noise:TEMPORAL\_SHIFT, noise:CROSS\_CATEGORY\_TRANSFER, value\_effectiveness:CONCENTRATE, value\_effectiveness:CROSS\_CATEGORY\_TRANSFER, value\_effectiveness:PERSONALISE

**evidence ids:** taxonomy:noise, RP-05, RP-07, TC-R11

**category assumptions:** A5, A6

**Field evidence behind these paths (titles)**

- Self-reported health impacts of do-it-yourself air cleaner use in a smoke-impacted community
- Real-World Effectiveness of Portable Air Cleaners in Reducing Home Particulate Matter Concentrations
- Reduction of personal PM2.5 exposure via indoor air filtration systems in Detroit: an intervention study
- The impact of portable air cleaners on indoor particulate matter concentrations and perceptions of indoor air quality: a randomized crossover trial in three multifamily buildings
- Analysis of sources and sinks of indoor particulate matter reveals insights into the real-world efficacy of portable air cleaners in a randomized intervention trial
- Indoor environmental health and asthma relief: findings from a longitudinal HEPA filter trial
- Effectiveness of portable HEPA air cleaners on reducing indoor PM2.5 and NH3 in an agricultural cohort of children with asthma: A randomized intervention trial

**Design DNA**

**F consumer friction (taxonomy:noise)** Present

Real Amazon review-text friction theme (see Consumer Pain methodology).

**O design operator** Method choice

Authored operator vocabulary applied deterministically (METHOD\_CHOICE, not evidence) - see OPERATORS/THEME\_OPERATORS in src/real/magic\_box\_real.py.

**S signal (noise)** Present

Motor/fan noise is a recurring complaint, now backed by peer-reviewed evidence it drives real under-use.

**T scientific tension** Present

Filtration vs. noise-driven adherence; Constant vs. automatic operation; CADR theory vs. real-home performance; Pollutant reduction vs. clinical outcome certainty

**R rival gap** Present

11 named real competitors measurably weaker on this theme. [Dyson, Plixio, Oreck, Envion, Wynd, Molekule, AMEIFU, AZEUS, AIRTHEREAL, Egofine, Elechomes]

**C versuni capability** Missing / unverified

No real Versuni internal capability/org-readiness dataset exists in this pipeline.

**A category assumption** Present

Clean-air delivery rate (CADR) is the primary performance basis.; Continuous, constant operation is the default expectation.

**E economic condition** Present

Real price-weighted exposure computed for this theme (\$84,015.62).

## 5. Comparable products

**comparable set:** 10 verified official Versuni/Philips air-purifier families (room-scale)  
**clean-air delivery rate (CADR):** 165-600 m3/h across 9 real comparables  
source: individually verified official Versuni/Philips product pages (data/visual/product\_images.json)  
**room coverage:** 42-156 m2 across 9 real comparables  
source: individually verified official Versuni/Philips product pages (data/visual/product\_images.json)  
**minimum noise:** 13-20.5 dBA across 6 real comparables  
source: individually verified official Versuni/Philips product pages (data/visual/product\_images.json)

## 6. Engineering envelope

| metric                         | value from verified comparables                                              | epistemic type      |
|--------------------------------|------------------------------------------------------------------------------|---------------------|
| clean-air delivery rate (CADR) | 165-600 m3/h (n=9 comparables)                                               | Observed comparable |
| room coverage                  | 42-156 m2 (n=9 comparables)                                                  | Observed comparable |
| minimum noise                  | 13-20.5 dBA (n=6 comparables)                                                | Observed comparable |
| target mass                    | No verified comparable publishes this value - left unknown, never estimated. | Unknown             |
| target power                   | No verified comparable publishes this value - left unknown, never estimated. | Unknown             |
| target dimensions              | No verified comparable publishes this value - left unknown, never estimated. | Unknown             |

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## 7. Economics

**price-weighted exposure:** \$84015.62

SUM of real observed listed prices across affected reviews with a known price. This is a RELATIVE exposure indicator, not a revenue or market-size estimate - it has no units-sold, conversion-rate, or time-period basis. It answers 'which friction touches more expensive products' not 'how much money is at stake per year.'

**comparable market median:** \$159.9 (n=35 products) - not this concept's price

MEDIAN real listed price across the 35 distinct real products affected by this friction that have a known price. This is what products in this segment actually cost today - not a proposed price for a new concept, which this evidence cannot establish.

**target price:** Not modelled - No concept target-price model exists - only the comparable market median (\$159.9, 35 real products) as reference. A target price would be invented.

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## 8. Prototype

**State:** [Concept visual](#)

Concept visual means a schematic exists, nothing more: no digital model, no physical prototype, no test data.

Machine-composed schematic from the innovation's stored fields only (operator rule, friction stats, comparable envelope). Not a design rendering; a Fable render or drawing would also be CONCEPT\_VISUAL, never a digital model or physical prototype.

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## 9. Experiment and next learning

### Next experiment

A real acoustic-engineering standard or published low-RPM motor precedent, if found, would raise this to high; a real internal R&D estimate showing it requires novel hardware would lower it to low.

### Kill criterion

No authored kill criterion yet for this innovation - only the formal case (see formal\_case.abandon\_signal) has a human-authored one; this innovation earns its own by developing further evidence or by a human decision to invest.

10. What could kill it?

Where this idea stands: Developing

Non-dominated and grounded in a real research tension.

**kill criterion:** No authored kill criterion yet for this innovation - only the formal case (see formal\_case.abandon\_signal) has a human-authored one; this innovation earns its own by developing further evidence or by a human decision to invest.

Uncertainties

- No real acoustic-engineering trend document exists in this corpus at all - the medium rating is a judgment call made in the absence of any external precedent, not a citation.

Critic dimensions

|             |                |                                                                                                            |
|-------------|----------------|------------------------------------------------------------------------------------------------------------|
| Evidence    | Survive        | Real Consumer Pain evidence: prevalence 3.5%, average rating gap -0.571 - materially above the gate floor. |
| Economic    | Survive        | Real price-weighted exposure (\$84,016) is at or above the corpus median (\$16,113).                       |
| Competitive | Survive        | Real white space: 11 named real rivals are measurably weaker on this theme.                                |
| Human       | Needs evidence | No real usability/human-factors evidence exists in this pipeline for this concept.                         |
| Physical    | Needs evidence | No real lab/engineering feasibility data exists in this pipeline for this concept.                         |
| Versuni fit | Needs evidence | No real organizational-capability assessment exists in this pipeline for this concept.                     |
| Timing      | Needs evidence | No real market-timing/window evidence exists in this pipeline for this concept.                            |
| Robustness  | Needs evidence | No real sensitivity/robustness study exists in this pipeline for this concept.                             |

**critic overall:** Needs evidence

Criteria results

Needs evidence: 32 - Pass: 13 - Challenge: 4 - N/A: 3

|    |           |                                                                                                                                                                    |
|----|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| V8 | Challenge | Operator AMBIENT may add surface area to manage - unverified.                                                                                                      |
| B5 | Challenge | Filtration vs. noise-driven adherence; Constant vs. automatic operation; CADR theory vs. real-home performance; Pollutant reduction vs. clinical outcome certainty |
| P2 | Challenge | Filtration vs. noise-driven adherence; Constant vs. automatic operation; CADR theory vs. real-home performance; Pollutant reduction vs. clinical outcome certainty |
| F3 | Challenge | medium                                                                                                                                                             |

Lineage and provenance

**generated by:** src/real/innovations\_real.py  
**magic run input sha256:** 0b3f9862e0e270a295eacd709182802e981090e817f4f1072000e37aa83f150c  
**funnel snapshot:** 933035cfaeb5c7adef564b21e11d2ef4633b1f7328d25088269a198a80cfd15c  
**figure source:** /concept-visuals/noise\_AMBIENT.svg (generated 2026-09-01T08:57:20.332044+00:00 by Claude Fable 5 generator - src/real/concept\_visuals.py)  
**data source:** data/processed/innovations\_real.json + data/processed/funnel\_real.json (deterministic; no invented fields)  
**dossier generator:** src/real/generate\_innovation\_dossiers.py